Weekly Test

NCERT IV - MATHS 211

Roll No. | | | | | | | | | | | | |

Code No. SCIENCE/WT1/X/26
Set A

Day and Date of Examination ______________________________

Signature of Invigilat1ors 1. ______________________________

2. ______________________________

General Instructions:

1. Candidate must write his/her Roll Number on the first page of the Question Paper.

2. Please check the Question Paper to verify that the total pages and the total number of questions
contained in the Question Paper are the same as those printed on the top of the first page. Also check to see that the questions are in sequential order.

3. For the objective-type of questions, you have to choose any one of the four alternatives given in the question i.e. (A), (B), (C) or (D) and indicate your correct answer in the Answer-Book given to you.

4. All the questions including objective-type questions are to be answered within the allotted time and no separate time limit is fixed for answering objective-type questions.

5. Making any identification mark in the Answer-Book or writing Roll Number anywhere other than the specified places will lead to disqualification of the candidate.

6. In case of any doubt or confusion in the question paper, the English Version will prevail.

7. Write your Question Paper Code No. SCI/WT1/X/26, Set A on the Answer-Book.

8. (a) The Question Paper is in English/Hindi medium only. However, if you wish, you can answer in any one of the languages listed below:
English, Hindi, Urdu, Punjabi, Bengali, Tamil, Malayalam, Kannada, Telugu, Marathi, Odia, Gujarati, Konkani, Manipuri, Assamese, Nepali, Kashmiri, Sanskrit and Sindhi.
You are required to indicate the language you have chosen to answer in the box provided in the Answer-Book.

(b) If you choose to write the answer in the language other than Hindi and English, the responsibility for any errors/mistakes in understanding the questions will be yours only.

General Instruction:

1. Answers of all questions are to be given in the Answer-Book given to you.
2. 15 minutes time has been allotted to read this Question Paper. The question paper will be distributed at 15 minutes before; the students will read the question paper only and will not write any answer on the Answer-Book during this period.

Note:

(i) This question paper consists of 25 questions in all.
(ii) All questions are compulsory.
(iii) Marks are given against each question.
(iv) Section – A consists of:
(a) Q. No. 1 to 10 or 1 to 17 – Multiple Choice type Questions (MCQs) carrying 1 mark
each. Select and write the most appropriate option out of the four options given in each of these questions.
(b) Q. No. 11 to 15 or 18 to 28 – Objective type questions. Q. No. 11 to 15 or 18 to 27 carry 2 marks each (with 2 sub-parts of 1 mark each) and Q. No. 16 or 28 carries 5 marks (with 5 sub-parts of 1 mark each). Attempt these questions as per the instructions given for each of the questions 11 to 15 or 18 to 28.
(v) Section – B consists of:
(a) Q. No. 17 to 21 or 29 to 37 – Very Short Answer type questions carrying 2 marks each.
(b) Q. No. 22 to 23 or 38 to 42 – Short Answer type questions carrying 3 marks each.
(c) Q. No. 24 or 43 to 44 – Long Answer type questions carrying 5 marks each.

Time : 2 Hours 30 Mins **Maximum Marks:** 85

	SECTION A	
	1. Which of the following is not an SI unit? (A) meter (B) pound (C) kilogram (D) second	1
	2. A mass of $10\mu g$ is equivalent to: (A) 10^{-6}gm (B) 10^{-3}gm (C) 10^{-9}gm (D) 10^{-12}gm	1
	3. Which subatomic particle is not found in the nucleus and has a much lower mass than the others? (A) Proton (B) Neutron (C) Electron (D) Alpha	1

	4. Which of the following subatomic particles is usually found in the nucleus of an atom? (A) Protons and neutrons only (B) Protons, neutrons, and electrons (C) Neutrons only (D) Electrons and neutrons only	1
	5. According to the periodic law given by Mendeleev, the properties of an element are a periodic function of its: (A) Atomic volume (B) Atomic size (C) Atomic number (D) Atomic mass	1
	6. The number of elements present in the 5th period of the modern periodic table is: (A) 2 (B) 8 (C) 18 (D) 32	1
	7. Which of the following compounds contains an ionic bond? (A) Hydrogen chloride (B) Calcium chloride (C) Oxygen gas (D) Nitrogen gas	1

	8. Covalent compounds generally have lower melting points than ionic compounds because: (A) They are always soluble in water (B) They contain free-moving ions in the solid state (C) The forces of attraction between their molecules are weak (D) They consist of charged species held by strong electrostatic forces	1
	9. Which of the following is not a strong acid? (A) Hydrochloric acid (HCl) (B) Hydrobromic acid (HBr) (C) Hydroiodic acid (HI) (D) Hydrofluoric acid (HF)	1
	10. Self-dissociation of water produces: (A) A large number of H^+ ions (B) A large number of OH^+ ions (C) H^+ and OH^+ ions in equal numbers (D) H^+ and OH^+ ions in unequal numbers	1
	11. The type of friction between two surfaces before an object starts moving is called: (A) Sliding friction (B) Rolling friction (C) Static friction (D) Fluid friction	1

	12. How does the gravitational force between two objects change if the distance between them is doubled? (A) It doubles (B) It quadruples (C) It reduces to one-fourth (D) It reduces to half	1
	13. The acceleration due to gravity (g) on the surface of the earth is: (A) Directly proportional to the mass of the falling body (B) Independent of the mass of the falling body (C) Greater at the equator than at the poles (D) Equal to zero everywhere on the surface	1
	14. A body of mass 0.5 kg possesses 100 J of kinetic energy. What is the velocity of the body? (A) 10 m/s (B) 20 m/s (C) 200 m/s (D) 400 m/s	1
	15. Two bodies of equal mass move with uniform velocities u and $4u$ respectively. What is the ratio of their kinetic energies? (A) 1:2 (B) 1:4 (C) 1:8 (D) 1:16	1

	16. In which of the following media will the speed of light be the maximum? (A) Medium A (Refractive index = 1.6) (B) Medium B (Refractive index = 1.3) (C) Medium C (Refractive index = 1.5) (D) Medium C (Refractive index = 1.4)	1
	17. Which of the following quantities remains constant during the reflection of light? (A) Speed of light (B) Frequency of light (C) Wavelength of light (D) All of the Above	1
	Write TRUE for correct statement and FALSE for incorrect statement	
	18.	
	i. $1\,mm^2 = 10^{-3} m^2$	1
	ii. Electrons radiate energy continuously while moving in stationary orbits.	1
	19.	
	i. The vertical columns in the periodic table are called periods.	1
	ii. Covalent compounds are good conductors of electricity in aqueous solutions.	1

	20.	
	i. Aqueous solutions of all salts are neutral in nature (i.e., neither acidic nor basic).	1
	ii. Balanced forces acting on an object can change its state of rest or uniform motion.	1
	21.	
	i. According to the law of conservation of energy, energy can neither be created nor destroyed during transformation.	1
	ii. When a ray of light enters from a rarer medium to a denser medium, its speed decreases.	1
	Fill in the blanks	
	22.	
	i. The SI unit of luminous intensity is _______.	1
	ii. The maximum number of electrons that can be accommodated in the L-shell (n=2) is _______.	1
	23.	
	i. According to the modern periodic law, the properties of elements are a periodic function of their _______.	1
	ii. An oxygen molecule O_2 contains a total of _______ shared pairs of electrons between its two atoms.	1
	24.	
	i. Acids taste _______, while bases taste _______.	1
	ii. The force acting perpendicular to the surface of an object is called _______.	1

	25.	
	i. When a body is fully or partially immersed in a fluid, it experiences an upward force equal to the _______ of the fluid displaced by it.	1
	ii. The constant of proportionality G in Newton's law of gravitation is called the _______ gravitational constant.	1
	26.	
	i. The rate of doing work is defined as _______, and its SI unit is the watt.	1
	ii. If the height of an object above the ground is halved, its potential energy becomes _______ of its initial value.	1
	27.	
	i. The phenomenon of splitting white light into its constituent seven colors when passing through a prism is called dispersion of light.	1
	ii. Short-sightedness (myopia) can be corrected by using a concave (diverging) lens.	1
	28.Match the columns	5

	Column A		Column B
A	Object at $20\,cm, f = -12\,cm$ concave mirror)	1	Magnification is $+2$ (virtual, erect)
B	Object at $10\,cm, f = +20\,cm$ convex mirror)	2	Image formed at $-30\,cm$ (real, inverted)
C	Object at $10\,cm, f = -20\,cm$ concave mirror)	3	Image formed at $+6.67\,cm$ (virtual, erect)
D	Half covered Convex lens	4	Complete image with reduced brightness
E	Correction for Myopia	5	Cylindrical Lens
		6	Concave lens

	29. Name the SI units used while purchasing silk ribbon, milk, and potatoes.	2
	30. What does the atomic number tell us about the composition of a neutral atom?	2
	31. Name the fundamental property of an element on which the modern periodic law is based.	2
	32. Name the type of chemical bonds present in a molecule of water (H_2O).	2
	33. What chemical causes a severe pain and burning sensation when someone accidentally touches a nettle plant?	2
	34. Calculate the momentum of a body of mass $10\,Kg$ moving with a velocity of $7\,m\,s^{-1}$.	2
	35. What is the mass of an object whose weight is $49\,N$? $(Take\ g = 9.8\,m\,s^{-2})$	2
	36. Calculate the work done in lifting a body of mass $100\,kg$ vertically through a height of $10\,m$. $(Take\ g = 10\,m\,s^{-2})$	2
	37. What is the cause of dispersion of light when it passes through a glass prism?	2
	38. Identify the common unit used to measure human body temperature and state its corresponding SI unit. Or Explain what is meant by a "stationary state" in Neils Bohr's atomic model.	3

	39. State Dobereiner's law of triads and show how chlorine, bromine, and iodine (atomic masses 35.5, 80, and 127 respectively) form a triad. Or Explain the step-by-step formation of a sodium ion (Na^{+}) from a neutral sodium atom (Na).	3
	40. What are indicators? Mention the colour change observed when methyl orange is added to (i) an acidic medium (ii) a basic medium. Or State Newton's first law of motion. Why do standing passengers in a stationary bus fall backward when the bus starts moving suddenly?	3
	41. A stone is dropped from the top of a $45\,m$ high tower. Calculate its velocity just before it hits the ground. ($Take\ g = 9.8\,m\,s^{-2}$) Or State Archimedes' principle. Explain why a capped empty plastic bottle pushed under water bounces back to the surface when released, and list two practical applications of Archimedes' principle in daily life or industry.	3

	42. Define work and power. Write their respective SI units. Or An object is placed at a distance of 20 cm in front of a concave mirror of focal length 12 cm. Find the position of the image formed and calculate its linear magnification.	3
	43. i) State three characteristic differences between ionic compounds and covalent compounds with respect to their physical state, melting point, and electrical conductivity. Or ii) Describe Ernest Rutherford's alpha particle scattering experiment using gold foil. Explain how the results led to the discovery of the atomic nucleus and state the main features of Rutherford's model.	5
	44. i) Define pH. Calculate the pH of a $1.0\,X\,10^{-4}\,M$ solution of nitric acid (HNO_3) and state whether the solution is acidic, basic, or neutral. Or ii) A block of wood kept on a table exerts a force (thrust) of $49\,N$. The dimensions of the block are $40\,cm\,X\,20\,cm\,X\,10\,cm.$ Calculate the pressure exerted by the block on the table top when it lies on its side with dimensions: (a) $20\,cm\,X\,10\,cm$ (b) $40\,cm\,X\,20\,cm$	5